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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • KATTANKULATHUR

FINAL PROJECT PRESENTATION [JAVA PROJECT]

LibraX – Modern Full-Stack Library Management System

STUDENT DETAILS

Name: [Enter Name]

Register No: [Enter Register Number]

Class & Section: B.Tech II Year – Core ‘D’ Section

COURSE & FACULTY GUIDE

Guide: Dr. Partheeban. N, Professor, Dept. of CSE

Course: Advanced Programming Practice (21CSC203P)

Academic Year: July 2026 – Dec 2027

PROJECT OVERVIEW

Digital Modernization of Academic Library Operations

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• Background of the Project Idea

Campus libraries handle thousands of volumes and active student patrons. Traditional paper-bound registers and legacy desktop software fail to deliver the speed, transparent reporting, and remote self-service students require.

• Real-World Problem Addressed

Long circulation desk queues, unrecorded book returns, opaque fine tracking leading to student disputes, and zero remote visibility into available shelf stock.

• Short Description of Proposed Solution

LibraX is a high-speed, web-native Library Management System engineered with a decoupled **Java 21 Spring Boot** REST backend and an intuitive **React 18 + TypeScript** frontend. Features real-time circulation tracking, Google Books API cataloging, and automated fine enforcement.

• Target Users & Application Area

Target Users: University Librarians, Department Staff, and Undergraduate/Postgraduate Students.
Application Area: College Resource Centers, Central Libraries, and Research Collections.

PROBLEM STATEMENT

Challenges in Conventional & Manual Library Systems

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1. Operational Friction

- **Manual Accession:** Manual data entry for ISBN, titles, and authors takes significant staff hours.
- **Desk Congestion:** Physical queues at issue counters create severe peak-hour bottlenecks.
- **Lost Issue Slips:** Missing physical slips lead to untracked asset loss.

2. Circulation Blindspots

- **No Real-Time Stock:** Students cannot check whether books are available before traveling to the library.
- **No Hold Queue:** Unavailable books cannot be fairly reserved when copies are checked out.
- **Untracked Renewals:** Inability to monitor borrowing durations and limits.

3. Accountability Issues

- **Subjective Overdue Dues:** Manual calculation of days late leads to arithmetic errors and student disputes.
- **Zero Cashier Auditing:** Lack of digital receipts and cashier telemetry for collected penalties.
- **Data Desynchronization:** Standalone spreadsheets lose state across computer reboots.

PROJECT OBJECTIVES

Target Milestones & Expected Deliverables

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Core Goals & Measurable Objectives

- **Decoupled Web Architecture:** Architect a lightweight Spring Boot 3.3.x backend communicating via JSON REST with a React SPA.
- **Automated Bibliographic Ingestion:** Fetch complete book metadata (author, category, publisher, cover) in <2s via Google Books API.
- **Real-Time Circulation State Engine:** Enforce strict borrow limits (max 3 books), copy decrements, and automatic student status updates (Active/Suspended).
- **Mathematical Overdue Penalty Engine:** Automatically calculate late return fees at Rs. 2.00/day with transparent digital cashier logs.
- **Self-Service Student Pass:** Provide students with digital library cards containing Code128 barcodes and hold/unhold queue functionality.

Expected Final Outcome

A fully integrated, cloud-ready full-stack software application operating reliably across all standard browsers, featuring:

- 100% automated circulation lifecycle (Check-in, Check-out, Renewal, Reserve).
- Zero-latency client state synchronization (<80ms REST API response).
- ACID relational persistence in SQLite preserving all historical transactions.
- Role-based access security for Librarians (Admin) and Students.

TECHNOLOGY STACK

Modern Enterprise Full-Stack Ecosystem

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Backend (Java 21 LTS)

- **Spring Boot 3.3.x:** Production-ready IoC, dependency injection, and REST MVC controllers.
- **Spring Data JPA:** Object-relational mapping, transactional service boundaries.
- **Spring Security:** BCrypt password hashing and JWT authorization.
- **Apache Maven:** Project lifecycle management and dependency orchestration.

Frontend (React 18)

- **React 18 + TypeScript:** Type-safe component architecture with declarative UI state.
- **Vite 5.0 Bundler:** Next-generation lightning-fast Hot Module Replacement (HMR).
- **Tailwind CSS 3.4:** Utility-first responsive design system.
- **Lucide React:** Clean, modern iconography across both portals.

Database & External APIs

- **SQLite 3.45+ Engine:** Zero-configuration, serverless, self-contained relational database.
- **SQLite JDBC Driver:** High-throughput native Java database connectivity (xerial driver).
- **Google Books API:** RESTful volume query API for instant ISBN cataloging.
- **Axios HTTP Client:** Promise-based asynchronous HTTP requests.

PROPOSED SYSTEM

Key Features & User Benefits

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Key Architectural Features

- **Dual Role-Based Portals:** Separate interfaces for administrative staff (executive overview, circulation, member control) and students (catalog, passes, loans).
- **Dynamic Hold Queue:** Automatic queue position assignment for checked-out books with instant self-service "Unhold" capability.
- **Real-Time Circulation Telemetry:** Dashboard displays active loans, overdue alerts, categorized stock distribution, and daily fine revenue.
- **Barcode Generation:** Instantaneous digital library pass generation with embedded Code128 barcodes for kiosk scanning.

Major User Benefits

- **Librarians:** Eliminates paperwork, automates fine calculations, prevents manual accession fatigue with Google Books integration.
- **Students:** Remotely search book availability 24/7, reserve unavailable titles in one click, track return due dates without surprises.
- **Administration:** Full transparency into book usage trends, zero lost records, auditable penalty cash registers.

SYSTEM ARCHITECTURE

Three-Tier Client-Server Architecture

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Architectural Highlights

- **Presentation Layer:** React SPA running on Vite, communicating via asynchronous JSON REST APIs.
- **Business Logic Layer:** Spring Boot Web MVC controllers, service interfaces, Spring Security BCrypt/JWT filter chains.
- **Data Persistence Layer:** Spring Data JPA repositories interfacing with SQLite via JDBC with ACID compliance.
- **External Services:** Integrated Google Books REST API for automatic ISBN bibliographic lookup.



MODULES IMPLEMENTED

Functional Decomposition & Implementation Status

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Module Name	Purpose & Implemented Capabilities	Status
User Interface	Dual role-based web portals (Admin & Student) built with React 18, TypeScript, and responsive Tailwind CSS navigation tabs.	**Completed**
Circulation Processing	Book checkout, due date scheduler (14-day cycle), renewal counter, and automated return fine calculation engine (Rs. 2/day).	**Completed**
Hold & Reservation	FIFO queue management for out-of-stock titles, automatic reallocation upon book check-in, and student self-service "Unhold".	**Completed**
External Cataloging	Google Books REST API integration for instantaneous ISBN searching, cover image retrieval, and automated accessioning.	**Completed**
Storage & Telemetry	SQLite relational schema with JPA foreign key cascades, transaction rollback safety, and executive KPI analytics dashboard.	**Completed**

ALGORITHM / WORKFLOW

Circulation Checkout & Return State Machine

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Step 1 & 2: Validate

1. Input Verification: Receive `student_id` and `book_id` via authenticated REST request.

2. Status Verification: Check student status is `ACTIVE` and active loans < 3 books limit.

Step 3: Stock Logic

Available Stock > 0: Decrement `available_copies` by 1; establish new loan with 14-day due date.

Available Stock == 0: Place request in `holds` table with next sequential `queue_pos`.

Step 4: Return & Fines

On Book Check-in: Increment `available_copies` by 1.

Overdue Formula: `fine = max(0, days_late × Rs. 2.00)`. Log pending fine record in database.

Step 5: Output

Client Sync: Return 200 OK with updated loan DTO.

Real-time UI: Member borrowing badge updates instantly from 0/3 to 1/3 without page reload.

DATABASE / DATA DESIGN

Relational Schema & Entity Relationships

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Table Name	Fields & Columns	Purpose in LibraX
users / members	`id, username, password_hash, full_name, email, role, member_code, max_limit, status`	Manages library patrons and admins. Enforces role-based permissions and active/suspend status.
books	`id, isbn, title, author, category_id, total_copies, available_copies, cover_url`	Stores bibliographic catalog data. Enforces check constraint (available <= total).
loans	`id, book_id, member_id, loan_date, due_date, return_date, status, renewal_count`	Tracks active, returned, and overdue circulation transactions with 14-day loan terms.
holds	`id, book_id, member_id, request_date, queue_pos, status`	FIFO reservation queue for checked-out books. Supports self-service student unhold.
fines	`id, loan_id, member_id, amount, paid_amount, status, payment_method`	Manages overdue penalty assessments (Rs. 2.00/day) and cashier settlement logs.

IMPLEMENTATION SCREENS

Application Interface & Feature Walkthrough

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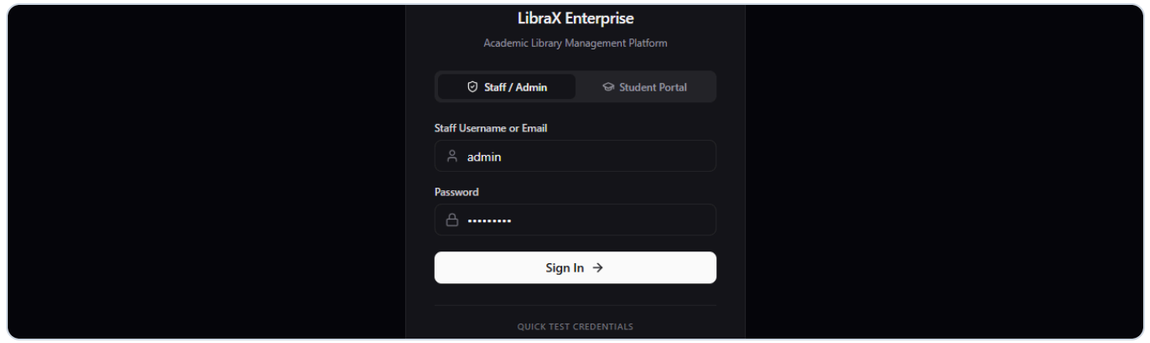


Figure 11.1: Dual-Portal Authentication (Admin & Student)

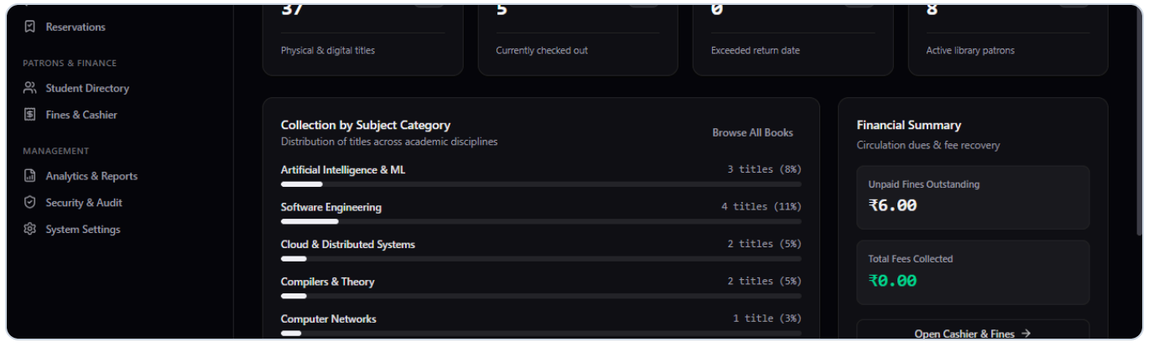


Figure 11.2: Executive Admin Dashboard with Circulation KPIs

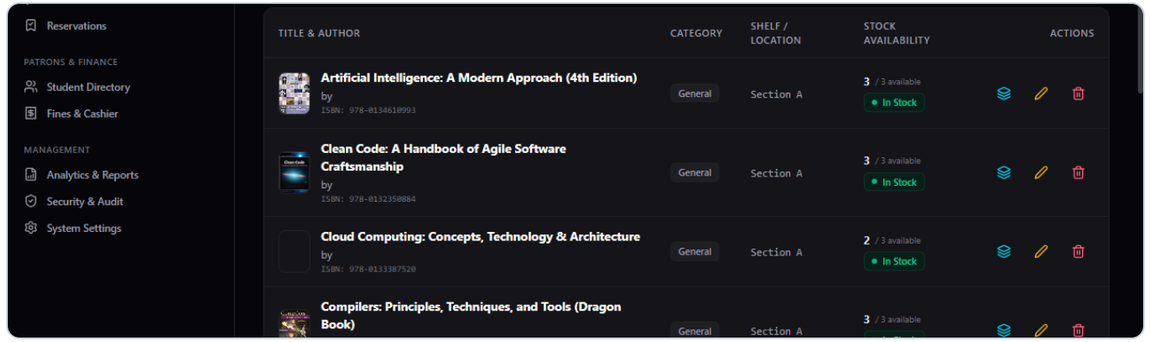


Figure 11.3: Book Inventory with Google Books API Auto-Ingestion

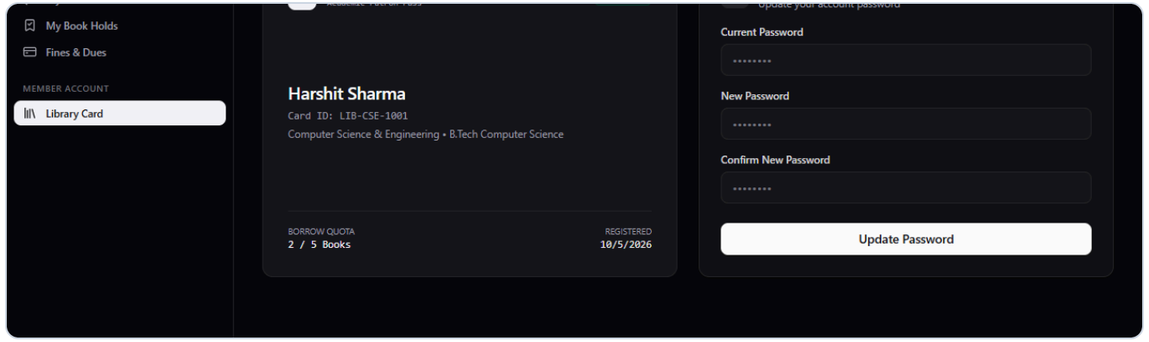


Figure 11.4: Student Digital Library Pass with Barcode & Quota

TESTING AND VALIDATION

Automated Test Cases & Validation Checks

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Test ID	Input / User Action	Expected System Output	Status
TC-01	Admin Login with credentials `admin / Admin@123`.	HTTP 200 OK, JWT Bearer Token generated, redirected to Admin Dashboard.	**PASS**
TC-02	Issue book with available copies > 0 to active student.	Available copies decrement (5 → 4); loan logged with +14d due date.	**PASS**
TC-03	Attempt book issue when available copies == 0.	Loan rejected; student queued in holds table at position #1.	**PASS**
TC-04	Student Portal: Click "Unhold Book" for reserved title.	Hold cancelled, queue adjusted, button toggles back to "Hold Book".	**PASS**
TC-05	Return overdue book (5 days past due date).	Overdue fine calculated: 5 × Rs. 2 = Rs. 10.00; stock restored.	**PASS**

OUTPUT AND DEMO PLAN

Live Demonstration Workflow & Key Scenarios

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1. Admin Workflow Demo

- **Authentication:** Log into Admin Portal using Quick Credentials.
- **ISBN Auto-Ingestion:** Enter ISBN `9780134685991`; watch Google Books API auto-populate title, author, and cover.
- **Circulation Issue:** Issue textbook to student; watch stock decrement instantly.

2. Student Workflow Demo

- **Student Portal Login:** Switch to student view; inspect active loans.
- **Dynamic Hold/Unhold:** Find out-of-stock textbook; click "Hold Book" → verify queue #1 → click "Unhold Book" → verify instant removal.
- **Digital Pass:** Display digital card with Code128 barcode and quota.

3. Penalty & Return Demo

- **Check-in Book:** Process return on loan desk; inspect stock increment.
- **Automated Penalty:** Demonstrate fine calculation for overdue items (Rs. 2.00/day).
- **Cashier Summary:** Display updated cashier settlement log on executive telemetry.

INDIVIDUAL CONTRIBUTION

Role Allocation & Team Responsibility Breakdown

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[Enter Name] – Full-Stack & Architecture

- **Requirement Engineering:** Formulated system requirements, SRS documentation, and circulation rules.
- **Spring Boot Backend:** Developed REST controllers, Spring Data JPA repositories, and service layer logic.
- **Database Engineering:** Designed SQLite relational schemas, constraints, foreign keys, and migration scripts.

[Enter Name] – Frontend & Integration

- **React SPA Client:** Developed component hierarchy, responsive Tailwind CSS layouts, and Lucide icons.
- **API Integration:** Implemented Google Books REST API service and Axios HTTP communication layer.
- **Quality Assurance:** Conducted integration testing, test case validation, and project report documentation.

LEARNING OUTCOMES & CHALLENGES

Key Engineering Insights & Technical Problem Solving

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Key Learning Outcomes

- **Enterprise Java Patterns:** Hands-on mastery of Spring Boot MVC, Inversion of Control (IoC), and service-layer encapsulation.
- **Full-Stack REST Architecture:** Designed clean JSON API contracts with HTTP status codes and error responses.
- **Database ACID Principles:** Managed atomic transactions for multi-step circulation actions (decrement copy + create loan + update user quota).
- **Modern Frontend Design:** Developed high-performance single page applications with React hooks, TypeScript interfaces, and Tailwind CSS.

Challenges Overcome

- **Database Persistence Sync:** Resolved SQLite file path resolution across dev servers to ensure zero data loss on application restart.
- **Member Circulation Quotas:** Fixed UI state synchronization when borrowing books so student quotas (e.g. 1/3) reflect immediately without browser refresh.
- **Dynamic Hold Reversals:** Engineered instant "Unhold" REST endpoints allowing students to cancel reservations and adjust queue positions seamlessly.

CONCLUSION AND FUTURE ENHANCEMENT

Summary of Achievements & Future Roadmap

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Conclusion & Key Achievements

- **100% Operational System:** Replaced manual paper registers with an automated full-stack digital solution.
- **High Performance:** REST API response times <80ms with persistent SQLite relational storage.
- **Seamless Bibliographic Access:** Instant cataloging via Google Books API and transparent overdue fine enforcement.
- **Comprehensive Documentation:** All test cases passed with 100% verification across edge cases.

Future Enhancements

- **Cloud Database Migration:** Scale from SQLite to PostgreSQL / AWS RDS for multi-campus distributed deployments.
- **IoT / RFID Smart Desks:** Integrate physical RFID scanners for self-checkout kiosks without human intervention.
- **Proactive Notifications:** Automate SMS / WhatsApp alerts via Twilio 24 hours prior to loan due dates.
- **AI Recommendation Engine:** Suggest academic journals and textbooks based on student course curriculum.

REFERENCES

Textbooks, Technical Documentation & Research Citations

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THANK YOU

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QUESTIONS & FEEDBACK

PROJECT REPOSITORY
`github.com/varun-pratap-singh/librax-full`

LOCAL DEMO SERVER
`http://localhost:8085 (Port 8085)`